

Week-8 Maths-1

L1 Limits & Continuity

$$\lim a_n \rightarrow a$$

as n increases, a_n comes closer to a .

- the limit of f at a from the left exists & equals M

$$\lim_{x \rightarrow a^-} f(x) = M \quad \left(\text{for right } x \rightarrow a^+ \right)$$

Eg. $f(x) = \lfloor x \rfloor \quad \lim_{x \rightarrow a^-} \lfloor x \rfloor = \lfloor a \rfloor$

- if M is same for left & right, then limit of f exists.
- f is continuous at a , if the limit of f at a exists & $\lim_{x \rightarrow a} f(x) = f(a)$.

$$f(a_n) \rightarrow f(a), \text{ when, } a_n \rightarrow a$$

- some rules.

1. $\lim_{x \rightarrow a} f(x) = F$ then $(f+g)(x) = F+G$
 $g(x) = G$

$$2. cF$$

$$3. (f - g)(x) = F - G$$

$$4. (fg)(x) = FG$$

$$5. (f/g)(x) \text{ if } G \neq 0, \text{ then } F/G.$$

$$6. \lim_{x \rightarrow a} (f)(x) = L, \quad \lim_{x \rightarrow a} g(x) = L$$

$$f(x) \leq h(x) \leq g(x), \text{ then}$$

$$\lim_{x \rightarrow a} h(x) = L$$

$$\text{Eg. } f(x) = 5x^3 + 0.45x^2 - 2x + 100$$

$$\lim_{x \rightarrow a} f(x) = 5a^3 + 0.45a^2 - 2a + 100$$

$$= f(a)$$

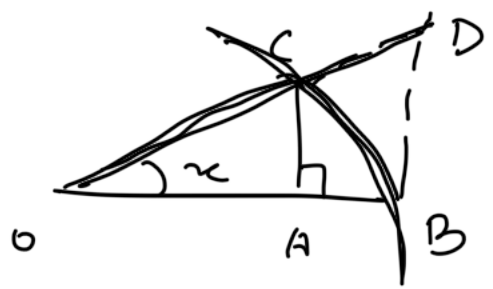
→ same proof for every polynomial

$$\text{Eg. } \lim_{x \rightarrow 0} \frac{5x^3 + 0.45x^2 - 2x + 100}{x^2 - 5x + 6} \rightarrow 100$$

6 $\Rightarrow$ Caution should check for roots

$$= \frac{\cancel{100}}{\cancel{6}} = \frac{50}{3} = 16.66$$

Eg. $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$



$$\frac{|BD|}{|OB|} = \tan x \quad |OB| = |OC| = 1$$

$$|AC| = \sin x$$

$$|BD| = \tan x \quad |\text{arc } BC| = x$$

$$\sin x \leq x$$

area of sector = $\frac{x}{2}$

$$\text{ar}(\triangle OBD) = \frac{1}{2} \times OB \times BD = \frac{\tan x}{2}$$

$$\frac{x}{2} \leq \frac{\tan x}{2} \Rightarrow x \leq \tan x$$

$$\sin x \leq x \leq \tan x$$

$$1 \leq \frac{x}{\sin x} \leq \frac{1}{\cos x}$$

$$1 \geq \frac{\sin x}{x} \geq \cos x \Rightarrow \frac{\sin(x)}{x} = 1$$

Eg. $\lim_{x \rightarrow 0} \frac{\tan(x)}{x} = \frac{\sin(x)}{x} \times \frac{1}{\cos(x)} = 1 \times 1 = 1$

Eg. $\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} = \frac{1}{2} \lim_{y \rightarrow 0} \left(\frac{\sin y}{y} \right)^2 = \frac{1}{2}$

Eg. $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = \left(\frac{1}{1+x} \leq \frac{\ln(1+x)}{x} \leq 1 \right) = 1$

- The funcⁿ f is said to be conti. if it is conti. at all pts. in its domain.

$$\lim_{x \rightarrow a} f(x) = f(a).$$
- Should be able to draw the graph w/o lifting our pencils.
- Egs. Polynomials, Rational funcⁿ with non-0 deno., e^x , $\log(x)$, $x > 0$, $\sin(x)$, $\cos(x)$.
- We talk abt. tangents only if the graph is continuous.

L2 Differentiability & Derivatives

- Recall - idea of rate of Δ .

$$\text{Speed} = \frac{\text{dist.}}{\text{time}} = \frac{2900}{72} \text{ km/hr} \approx 40 \frac{\text{km}}{\text{hr}}$$

or $\frac{260}{3} \approx 80+ \text{ km/hr}$
- Remember - Avg. & the instantaneous speeds are different.
- for instantaneous speed, we have to calculate the speed for a very short

time.

- Inst. speed = $\lim_{\Delta t \rightarrow 0} \frac{d}{t}$

- Let f be a funcⁿ defined on an open interval around a .

Then f is differentiable at a ,

if $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists.

Eg. $f(x) = x$ is differentiable at a pt. a .

Eg. $f(x) = \sin(x)$ is diff. at the pt 0 .

Eg. $f(x) = |x|$ & $f(x) = x^{1/3}$ is not differentiable at 0 .

Eg. $f(x) = \lfloor x \rfloor$ is not differentiable at any integer pt.

- If f is differentiable at a , then it is continuous at a .

$\lim_{x \rightarrow a} f(x)$, also, $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = L$

- $f(x) \Rightarrow$ derivative $f'(x)$

$$f'(x) = \frac{df}{dx}(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Its domain consists of those pts at which the funcⁿ $f(x)$ is differentiable

Eg. $f(x) = x$, $f'(x) = 1$

Eg. $f(x) = \sin(x)$, $f'(x) = \cos(x)$

- We will talk about tangents only if the funcⁿ is differentiable.

1.3 Computing Derivatives & L'Hopital's Rule

- if f is differentiable at a , then it is continuous at a .

- rules.

1. Linearity - $(f+g)'(a) = f'(a) + g'(a)$

2. $(cF)'(a) = cF'(a)$

3. $(f-g)'(a) = f'(a) - g'(a)$

4. Product rule

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a)$$

5. Quotient rule

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}$$

6. Composiⁿ / Chain rule

$$(f(g))'(x) = f'(g(x)) g'(x)$$

Eg. $f(x) = x^n ; n \in \mathbb{N}$
 $f'(x) = nx^{n-1}$ } you can prove this by inducⁿ.

Eg. $f(x) = 5x^3 - 17x^2 + \pi x - 0.5$
 $f'(x) = 15x^2 - 34x + \pi$

- for a polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

$$f'(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \dots + a_1$$

Eg. $f(x) = x^7 (\sin(x))$
 $f'(x) = 7x^6 \sin(x) + x^7 \cos(x)$

Eg. $f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$
 $f'(x) = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)}$
 $= \sec^2(x)$

Eg. $f(x) = \frac{1}{x^u} ; u > 0$

$$f'(x) = \frac{1 \times x^u - 1 \times u x^{u-1}}{(x^u)^2} = \frac{-u x^{u-1}}{x^{2u}}$$

$$= -x x^{-x-1}$$

eg. $f(x) = \tan(2x) = \frac{\sin(2x)}{\cos(2x)} = \frac{2\tan(x)}{1-\tan^2(x)}$

$$f'(x) = \sec^2(2x) \times 2 = 2\sec^2(2x)$$

- Indeterminate Limits - $f(x)$ & $g(x)$ are func^{ns}. Both limits are 0 or ∞ or $-\infty$. We want to compute -

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \Rightarrow \text{can't use quotient rule as it yields an indeterminate form.}$$

eg. $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = \frac{1}{1+x} = 1$

use L' Hopital's rule

- L' Hopital's rule

1. $f'(x)$ & $g'(x)$ exist on interval

2. $g'(x) \neq 0$ on this interval

3. $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$

then, $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$

$$\text{Eg. } \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x - 2} = \frac{2x - 5}{1} = -1$$

$$\text{Eg. } \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = \frac{\cos(x)}{1} = \cos(0) = 1$$

$$\text{Eg. } \lim_{x \rightarrow \infty} \frac{a + be^x}{c + de^x} = \lim_{x \rightarrow \infty} \frac{be^x}{de^x} = \frac{b}{d}$$

$$\text{Eg. } \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} = \frac{\sin(x)}{2x} = \frac{1}{2} \frac{\sin(x)}{x} = \frac{1}{2}$$

L4 Derivatives, Tangents & Linear Approximations

- recall differentiability $f'(x) = \frac{df}{dx}(x)$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- A tangent to $f(x)$ at a is a line which represents the instantaneous direcⁿ in which graph $\Gamma(f)$ moves at $(a, f(a))$.

- what is secant.



- tangent is basically a limit of secants.

$$y - f(a) = \frac{f(a+h) - f(a)}{a+h-a} (x-a)$$

is the eqⁿ of secant

- What happens in the limit?

tangent is given by,

$$y = f'(a)(x-a) + f(a)$$

If f is differentiable funcⁿ. Then tangent exists and vice-versa.

Eg. $f(x) = 5x^3 - 17x^2 + \pi x - 0.5$, $a = 0$

$$f'(x) = 15x^2 - 34x + \pi$$

$$f'(0) = \pi$$

$$\text{eq}^n \Rightarrow y = \pi(x-0) + f(0) = \pi x - 0.5$$

Eg. $f(x) = \cos(x)$, $a = \frac{\pi}{3}$

$$f'(x) = -\sin(x)$$

$$f'\left(\frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$\text{eq}^n \Rightarrow y = \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{3}\right) + \frac{1}{2}$$

Eg. $f(x) = x^{1/3} \Rightarrow$ not differentiable at 0.

$$\lim_{h \rightarrow 0} \frac{1}{h^{2/3}}$$

diverges to ∞ .
 $\therefore$ limit DNE

$y = mx + c$ works for lines which are not vertical.

- linear Approximation

$$L(x) = c + dx$$

$\Downarrow$

$$L(x) = f(a) + m(x-a)$$

we want to choose a linear funcⁿ which best approximates the funcⁿ $f(x)$ around pt. a .

$$f(x) \approx L(x) \quad \forall x \text{ close to } a.$$

- if $f(x)$ is differentiable at a ,

$$L(x) = f(a) + f'(a)(x-a)$$

and vice-versa.

best linear approximaⁿ.

Eg. $f(x) = x^3, \quad a = 1$

$$f'(x) = 3x = 3$$

$$L(x) = 3(x-1) + 1 = 3x - 2$$

Eg. $f(x) = \sec(x), \quad a = 0$

$$f'(0) = \tan(0) \sec(0) = 0$$

$$L(x) = 0(x-0) + \sec(0) \\ = 1$$